

Boundary-spanning interfaces between logistics, production, marketing and new product development

Edward A. Morash, Cornelia Dröge and Shawnee Vickery

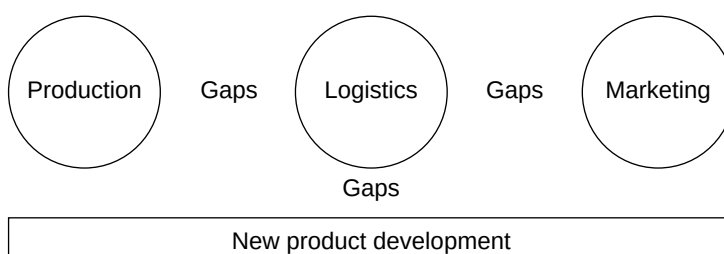
Introduction

Currently, it is fashionable to say that companies should be close to customers in meeting their needs and accommodating their unique requirements. Customer closeness and accommodation necessitate that everyone in the firm have the customer firmly in mind when exercising their functional capabilities. At the same time, functional capabilities are of necessity interdependent in creating value for customers. For example, logistics must work closely together with production and marketing to plan, co-ordinate and integrate their cross-functional activities. The modern way of expressing this is that logistics must serve both the internal customers and external customers of the firm[1].

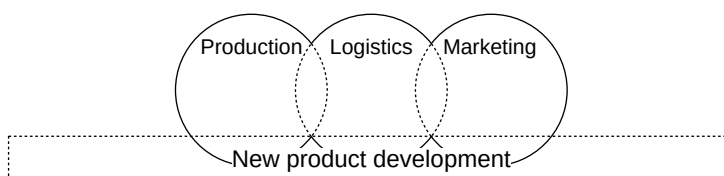
Customer accommodation and functional interdependencies contemplate process integration of firm capabilities. Too often a company's internal functions operate like functional "silos" rather than serving each other for the customer's benefit. As shown in the top portion of Figure 1, such isolated functionalism creates capability performance gaps which should translate into poor company performance. In contrast, the middle portion (B) of Figure 1 visually displays the appropriate and somewhat unique interface role of logistics with production, marketing, and new product development. As such, the current interest in process re-engineering and reinvention recognizes that all of these functional areas must work together co-operatively as unified processes for customer satisfaction and firm success[2-5]. Furthermore, logistics may be in an ideal interface position to facilitate this integration, as suggested by Figure 1.

Logistics is also unique in simultaneously interfacing with external suppliers, carrier partners and customers. As such, logistics is in a boundary-spanning role with these external "customers" as well[6-8]. The bottom portion (C) of Figure 1 is conceptually related to portion (B), since external "variety" and uncertainty must be internally matched and "absorbed" by internal cross-functional capabilities[9]. This has led some to suggest that logistics should take the primary interface responsibility for relationship management and supply chain integration[10].

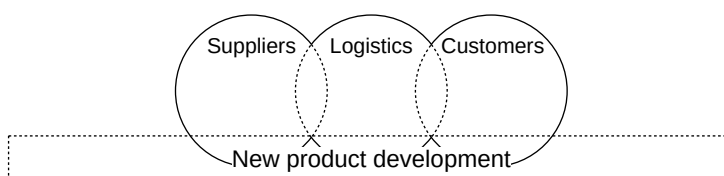
(a) Ineffective functional silos



(b) Interfunctional interfaces for internal boundary spanning



(c) Logistical interfaces for external boundary spanning

**Figure 1.**
Internal and external
functional relationships

The present research investigates empirically the cross-functional, interface, and overall business performance relationships for logistics, marketing, production and new product development. The next section will briefly discuss specific logistics boundary-spanning interface capabilities of customer service, logistics quality, channel distribution and total cost minimization. The methodology for investigating cross-functional, interface, and overall business performance relationships, will then be presented. This will be followed by the empirical results, a summary and discussion section and, finally, conclusions and implications.

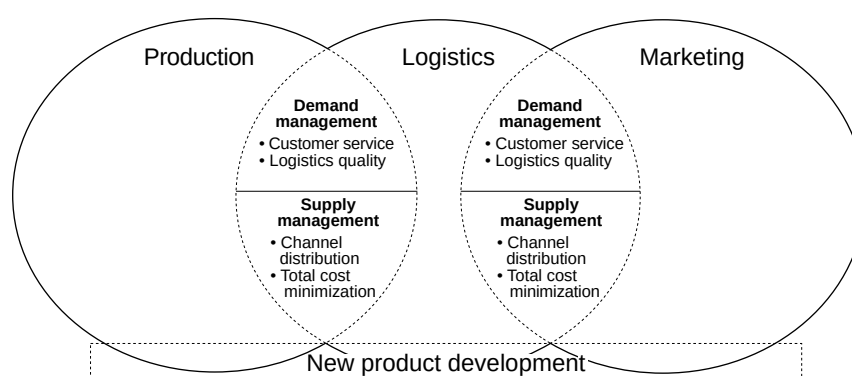
Demand interfaces versus supply interfaces

Recently, interest has grown in the demand management of logistics capabilities. Although logistics is more often thought of in terms of supply chain management and total cost minimization, it is sometimes suggested that

logistics can also differentiate product-service offerings[11], promote product demand, and “grow” a firm’s revenues[12]. It has been strongly asserted in some logistics literature that logistics customer service represents a major demand interface with the marketing mix[13] and that customer service capabilities can translate into market share gains[14]. More recently, logistics quality improvements have been enunciated as necessary conditions for world-class performance[15]. The term “demand-chain management” has recently been used to denote these demand-generating capabilities.

Figure 2 provides a visual representation of the demand and supply interfaces of logistics capabilities with the functional areas of production and marketing and the special “platform” case of new product development. Figure 2 is identical to the middle portion (B) of Figure 1, except that it shows the specific boundary-spanning interface capabilities investigated in this study for interfunctional process integration and overall firm performance. This study’s literature review identified logistics customer service, quality, channel distribution, and total cost minimization as major boundary-spanning interface capabilities. Although not meant to be exhaustive of logistics capabilities, these four concepts are mentioned most often in modern logistics literature and are central to logistics thinking. They also reflect, respectively, demand and supply chain management interface capabilities, which will be discussed next.

Figure 2.
Logistical interfaces and
capability links with
production, marketing,
and new product
development



Demand-management interface capabilities

Customer service is defined and measured in this study as the capability to be flexible and responsive in satisfying changing customer requirements and demands (e.g. requisite variety[9,16,17]). It is also measured as the ability to provide “core” services and facilitating or extra services which promote, differentiate, and facilitate the basic “core” product-service offering of the firm[18-20]. On the production side, and as shown in Figure 2, analogous customer service interfaces require that logistics be flexible and responsive in providing materials, components, and services to the internal customer of production. This is particularly true for demand-pull type systems such as JIT

(just-in-time) manufacturing, which explicitly adapts to customer demand as opposed to supply-push systems[6]. As such, the goal is cross-functional process integration or, more broadly, supply chain integration for overall firm performance.

Logistics quality is defined in terms of performance “gaps” and is measured as the ability to distribute a product or materials in conformance with customer requirements and standards, e.g. [15]. It is also measured as the ability to deliver products, materials and services without errors, defects, mistakes, or other gaps from customers’ expectations, where “customers” include both internal and external customers. For internal customers, logistics must facilitate and serve their functional capabilities in order to achieve process integration. Specific functional capabilities as used in this study for production, marketing, and new product development, are listed and defined in the Appendix Table AI. Although not exhaustive, these 23 capabilities are most representative of their respective functional disciplines.

Supply-management interface capabilities

Figure 2 also shows the supply interfaces of specific logistics capabilities with production, marketing and the “platform” case of new product development. In keeping with the logistics focus of this research, channel supply capabilities are defined and measured in terms of spatial, physical, and systemic properties. Channel distribution is measured as the ability intensively to distribute materials and products to selected target markets. It is also measured as the ability to extensively distribute to wide geographical areas[5,21].

In turn, logistics total cost minimization is measured as the capability to minimize total system costs so that cross-functional cost-tradeoffs are explicitly considered. It is also measured as process optimization, rather than functional suboptimization of isolated, individual functional capabilities. Together, channel distribution and total cost minimization make up the supply-management interface capabilities of logistics with production, marketing and new product development. This is also depicted in Figure 2.

Methodology

The present research is concerned with interfunctional process relationships, specific boundary-spanning interface relationships, and overall business performance relationships. To investigate these relationships, both importance and performance data were collected on three types of variables: business functions, specific logistics interface capabilities, and overall business performance indicators. Each type of variable will be discussed in turn.

Business function variables

The business functions investigated consisted of production, marketing, new product development and logistics. In each functional area, eight functional performance capabilities were identified which were especially representative and comprehensive of the respective function. This identification was based on

a thorough review of each functional discipline's literature. It was also facilitated by the research team being composed of resident expert specialists from each functional area. Further, subject respondents were asked to indicate which business function had primary responsibility for the capability, which was useful for checking classifications.

The eight functional measures for each area of production, marketing, and new product development are shown in the Appendix. This listing represents 23 specific functional capabilities (new product development actually had seven measures owing to its more narrow focus). The capabilities are also defined in Table AI. To reduce measurement error, these definitions were actually provided to each chief executive officer (CEO) subject respondent prior to surveying them. This ensured that all participants were operating from the same point of reference.

The functional capabilities were evaluated as to both importance and performance. As recommended by previous research, e.g. [15], both importance and performance data were collected at the same time. The subjects evaluated functional importance on a seven-point Likert-type scale ranging from 1 = least important for firm success, to 7 = most important for firm success, with a midpoint of 4. The functional performance evaluations for each functional capability were made on a seven-point Likert-type scale ranging from -3 = poor relative to major competitors, to +3 = excellent relative to major competitors, with a midpoint of 0. As recommended by previous research, functional performance was anchored relative to competitors; e.g. [15].

For each subject, the mean performance scores were calculated for each function. These functional means were then used as inputs for relationship testing involving inter-functional, interface, and overall business performance relationships.

Specific logistics interface variables

For logistics interface capabilities, the eight measures used are defined and discussed in the introduction. The measurement scales and data collection procedures were identical to the other previous business functions. These eight logistics interface capabilities measure the four underlying constructs of logistics customer service, logistics quality, channel distribution and logistics total cost minimization.

The four constructs were selected for several reasons. First, although not meant to be exhaustive of logistics capabilities, these four concepts are most often mentioned in logistics literature and are central to logistics thinking. Second, an important focus of the research is logistics boundary-spanning interfaces and integration with other business functions. These four concepts ideally cut across different functional areas and indeed, across different firm boundaries. Third, there was a desire to investigate relationships for both demand-management interfaces and supply-management interfaces. As discussed and defined in the introduction in some detail and as based on a comprehensive literature review (some of which is referenced at the end of this

article), customer service and logistics quality were identified as demand-management interfaces. In turn, channel distribution and logistics total cost minimization were identified as supply-management interfaces consistent with the introduction's definitions.

Eight different measures for these four logistics concepts were used for several reasons. First, there was a desire to be research symmetrical with the eight measures from each of the other functional areas (except for NPD). Second, at this point in the logistics discipline's evolution, some of the logistics concepts used are not precisely defined in the literature. It was therefore expected that repetitive measures of somewhat distinct definitions as supported by substantial literature, would improve construct validity and measurement reliability.

Business performance indicator variables

To measure overall business performance, data were collected on subject firms for five business performance indicators. The five financial performance ratios consisted of return on assets (ROA), return on investment (ROI), return on sales (ROS), and the two growth ratios of ROI growth and ROS growth. These five measures are very commonly used and mentioned in business financial and strategic performance research literature. All other logistics performance measures as used in this research are consistent with prior logistics performance literature; e.g. [22].

Subjects and sample

The subjects consisted of the CEOs from 65 furniture manufacturing firms. The furniture manufacturing industry was selected for several reasons. First (and possibly most important), this particular industry is highly competitive in that firms compete on a variety of functional and logistics interface capabilities. Second, the research team firmly believes that at this stage of theory building, single industry investigations have the advantage of controlling for industry effects. In research terminology, cross-industry variance is controlled. Thus, identified differences between firms are not due to differences in industry. Third, exogenous environmental influences are more likely to be identical for these firms. Finally, functional and interface performance can be anchored relative to major competitors in the same industry. This allows for cross-firm comparisons.

Survey instrument

The questionnaire was first pretested in the field with the CEOs of several furniture companies. Modifications to the instrument were made based on these interviews.

The questionnaire was then mailed to the CEOs of all US furniture firms which had more than ten million dollars in annual gross sales revenue. Soon after mailing the questionnaire, the CEO subjects were telephoned (a minimum of three times) to schedule a one to two-hour appointment time for them to

answer the questions over the phone. These personalized telephone interviews ensured that only the CEO, or another equally knowledgeable top executive, filled out the survey instrument. The research team believed that only the top executive would have the required broad-based knowledge which covered multiple functional areas of the firm. By ensuring the same title and position level for the respondents, position bias was minimized and cross-firm comparability was enhanced.

Personalized interviews were also used to increase the reliability and accuracy of the data, since respondents can be encouraged to ask questions and express any difficulties. As noted earlier, all concepts were specifically defined on the instrument itself (e.g. see the Appendix) to increase the study's validity and reliability. Thus, overall, the personalized interview procedure was expected to increase the quality of the data, but decrease participation somewhat owing to the time demanded. The final sample of 65 CEOs represented a response rate of approximately 20 per cent.

Stepwise regression

Since it was expected that some specific logistics interface variables might be intercorrelated, stepwise regression was selected as the most appropriate testing procedure for investigating their relationships to business performance indicators. Stepwise regression asks the question: "Which interface capability variable is most important in explaining overall business performance, and once that variance is accounted for, are there any other interface variables which are also significant predictors of performance?" For each model, stepwise regression first enters the independent variable which explains the most variation in the dependent performance variable. Additional independent variables are then entered one at a time only if they significantly explain additional "unique" variance in the dependent variable. At each step, variables already in the model are also re-evaluated to ensure that they are still significant or otherwise deleted. The documentation for this forward and backward stepwise regression procedure is in *SAS User's Guide: Statistics*, pp. 762-74. The level of tolerance is 0.15.

Results: importance of logistics compared to production, marketing and new product development

Table I compares the importance of logistics versus production, marketing and new product development, as rated by CEOs on a seven-point scale ranging from 1 = least important, to 7 = most important. As shown in Table I, logistics displays a significantly greater mean importance score ($\bar{X} = 5.62$) than all of the other functional areas. Consequently, CEOs view logistics as being more important for firm success than production ($\bar{X} = 5.34$), marketing ($\bar{X} = 5.09$) and new product development ($\bar{X} = 4.77$), in that descending order. However, it is important to note that all of these functional means are above the scale midpoint of 4.0, which likely implies some minimum acceptable level or industry standard to be a possible supplier.

(A) CEO importance ratings (1 to 7 scale, with 7, the most important; $\bar{X}$ = mean)^a

		Production ($\bar{X}$ = 5.34)	Marketing ($\bar{X}$ = 5.09)	New product development ($\bar{X}$ = 4.77)
Logistics importance:		<i>p</i> -values from 2-tailed tests ^b		
Overall logistics (log)	($\bar{X}$ = 5.62)	$p < 0.01$	$p < 0.01$	$p < 0.01$
Logistics quality	($\bar{X}$ = 6.11)	$p < 0.01$	$p < 0.01$	$p < 0.01$
Logistics customer service	($\bar{X}$ = 5.92)	$p < 0.01$	$p < 0.01$	$p < 0.01$
Channel distribution	($\bar{X}$ = 5.17)	$p = 0.32$	$p = 0.57$	$p = 0.04$
Logistics total cost minimum	($\bar{X}$ = 4.61)	$p < 0.01$	$p = 0.07$	$p = 0.53$

Notes: ^aImportance ratings were made on a 1- to 7-point scale from least important (1) to most important (7) with a midpoint of 4^b*p*-values are the statistical significance of the two-tailed comparison of two means; for example, *p*-values less than 0.01 indicate that the means are different at the 1 per cent level of significance**Table I.**
Functional comparisons
of the importance of
logistics versus
production, marketing
and new product
development

Considering the sub-categories of specific logistics capabilities, Table I also shows that logistics quality ($\bar{X}$ = 6.11) closely followed by customer service ($\bar{X}$ = 5.92) are rated by CEOs as significantly more important than production, marketing, and new product development. These differences with the functional means are all statistically significant at the $p < 0.01$ level. In contrast, the channel distribution variable ($\bar{X}$ = 5.17) is not significantly different from production and marketing and is only significantly more important than new product development. Further, total logistics cost minimization ($\bar{X}$ = 4.61) is significantly and interestingly less important than both production and marketing and not significantly different from new product development. Thus, in total, the demand-management capabilities of logistics quality and customer service are rated by CEOs as more important than the supply-management capabilities of channel distribution and logistics total cost minimization.

Functional performance relationships between logistics, production, marketing and new product development

Table II shows the performance intercorrelations between the functional areas of logistics, production, marketing and new product development, as well as with individual logistics capabilities. Separate functional performance measures are used in these relationship tests rather than the previous CEO importance ratings. The rationale is that what firms actually do is more germane to overall corporate performance than what CEOs say is important. However, in general, the importance and performance ratings are quite similar.

The top panel (A) of Table II shows that all of the functional areas are highly intercorrelated at the $p < 0.01$ level of significance. On average, logistics performance statistically “explains” approximately 16-25 per cent of the performance variation in the other functional areas. Thus logistics, production, marketing and new product development are all positively and significantly

Table II.
Functional performance
intercorrelations between
logistics, production,
marketing and new
product development;
and logistics interfaces

Performance ^a	Correlations (<i>p</i> -values)			
(A) Functional areas	(1) Logistics	(2) Production	(3) Marketing	(4) New product development
(1) Logistics	1.00			
(2) Production	0.45*** (<i>< 0.01</i>)	1.00		
(3) Marketing	0.49*** (<i>< 0.01</i>)	0.31*** (0.01)	1.00	
(4) New product development	0.42*** (<i>< 0.01</i>)	0.42*** (<i>< 0.01</i>)	0.43*** (<i>< 0.01</i>)	1.00
(B) Logistics interfaces				
Logistics quality	0.53*** (<i>< 0.01</i>)	0.36*** (<i>< 0.01</i>)	0.16 (0.22)	0.22* (0.08)
Logistics customer service	0.78*** (<i>< 0.01</i>)	0.51*** (<i>< 0.01</i>)	0.27** (0.03)	0.32*** (0.01)
Channel distribution	0.54*** (<i>< 0.01</i>)	0.11 (0.37)	0.47*** (<i>< 0.01</i>)	0.25** (0.04)
Logistics total cost minimum	0.58*** (<i>< 0.01</i>)	0.05 (0.69)	0.29** (0.02)	0.22* (0.07)

Notes: ^aPerformance evaluations were made on a –3 to +3 scale from poor to excellent relative to major competitors and with a midpoint of 0
*** ** * Significant at 0.01, 0.05, and 0.10, respectively (i.e. the correlation is significantly different from 0)

related to each other. This means that firms which perform well at logistics also tend to perform well with production, marketing and new product development. This is consistent with a process excellence orientation.

The two highest correlations in top panel (A) are between logistics and production and between logistics and marketing, rather than directly between production and marketing. These findings support the introduction's boundary spanning and interface roles for logistics, e.g. Figure 2. Further, new product development is almost uniformly correlated with each of the different functional areas. This supports its visual "platform" representation in the introduction's Figure 2.

The bottom portion (B) of Table II shows the intercorrelations between individual logistics capabilities and the different functional areas. Ignoring the first column of overall logistics performance with its component measures, the second column shows production performance related to individual logistics capabilities. Production performance is only significantly related to the demand management variables of logistics customer service followed by logistics quality. Comparing all functional areas, logistics quality is also most strongly related to

production performance. This finding is interesting and understandable since logistics quality is defined and concerned with “gaps” from specifications of customer requirements or from performance standards such as errors, mistakes and defects.

Possibly the most interesting and gratifying finding in panel (B), is that logistical customer service is most strongly related to production performance. This is also the highest correlation in the whole of Table II (except for logistics with its component measures). Logistics customer service is also significantly related to marketing performance, although at a lower level. These results are consistent with the discussion in the introduction as to logistics customer service representing a major demand management interface with both production and marketing, e.g. Figure 2.

Next, the third marketing column in panel (B) shows that the highest correlation for marketing performance is with the channel distribution variable. This finding is consistent with some logisticians' viewpoint that channels represent the “place” component of the marketing mix[5,21,23]. Furthermore, this finding supports the introduction's conceptualization that channel distribution is a major supply management interface, as depicted in Figure 2. This distinction between supply and demand management interfaces will be further discussed in the summary and discussion section.

The fourth column for new product development in panel (B) shows that all four logistics capabilities are correlated with it at approximately the same yet relatively modest levels. Again, this supports the viewpoint that new product development has a “platform” relationship with other functional areas. A final interesting finding in Table II relates to logistics total cost minimization. Logistics total cost minimization is not significantly related to production performance. In contrast, it is significantly related to marketing performance. The practical significance of this and other findings will subsequently be discussed in the summary and discussion section.

Relationships between functional area performance and overall business performance

Table III shows the relationships between individual functional area performance and five measures of overall firm performance. As discussed in the methodology section, these five business performance indicators include ROA, ROI, and ROS, and the two growth measures of ROI growth and ROS growth.

The top portion of Table III shows that out of the four functional areas, logistics is most strongly and consistently related to all of these business performance indicators. As such, logistics is positively and significantly related to ROA, ROI, ROS, ROI growth, and ROS growth. All five relationships are statistically significant at the $p \leq 0.01$ level.

In turn, production and new product development are related to some but not all of the five measures of overall firm performance; while marketing is not related to any performance indicator. However, this absence of significant

Performance	Correlations (<i>p</i> -values) for business indicators ^a				
(A) Functional areas	ROA	ROI	ROS	ROI growth	ROS growth
(1) Logistics	0.35*** (<i>< 0.01</i>)	0.32*** (0.01)	0.43*** (<i>< 0.01</i>)	0.39*** (<i>< 0.01</i>)	0.40*** (<i>< 0.01</i>)
(2) Production	0.25** (0.04)	0.28** (0.02)	0.19 (0.14)	0.20 (0.13)	0.04 (0.74)
(3) Marketing	0.14 (0.27)	0.14 (0.29)	0.13 (0.32)	0.14 (0.29)	0.01 (0.93)
(4) New product development	0.29*** (0.01)	0.35*** (0.01)	0.26** (0.04)	0.42*** (<i>< 0.01</i>)	0.09 (0.52)
(B) Logistics capabilities					
Logistics quality	0.19 (0.14)	0.21* (0.09)	0.30** (0.02)	0.33*** (0.01)	0.34*** (0.01)
Logistics customer service	0.36*** (<i>< 0.01</i>)	0.25** (0.05)	0.32*** (0.01)	0.31** (0.02)	0.33*** (0.01)
Channel distribution	0.09 (0.47)	0.19 (0.14)	0.21* (0.10)	0.17 (0.19)	0.13 (0.34)
Logistics total cost minimum	0.21* (0.09)	0.09 (0.49)	0.24* (0.07)	0.12 (0.37)	0.20 (0.14)

Table III.

Pearson correlations of logistics performance measures with business performance indicators

Notes: ^aBusiness performance indicators consist of the performance ratios Return on Assets (ROA), Return on Investment (ROI), Return on Sales (ROS), ROI growth, and ROS growth
*** ** * Significant at the 0.01, 0.05, and 0.10 levels, respectively

marketing relationships does not necessarily imply that marketing is generally not important. For example, prior research has shown marketing to be especially important for gains in market share, which is not tested here [14], or marketing may have lagged effects. Also, this particular industry may be a moderator.

The bottom portion of Table III shows the direct relationships between individual logistics capabilities and overall firm performance. As shown in panel (B) of Table III, customer service followed by logistics quality have the most consistently strong relationships with firm financial success. Customer service is positively related to all five success measures generally at the $p \leq 0.01$ level of significance. Logistics quality is significantly related to four of the measures and displays its strongest relationships with the two growth measures. In contrast, channel distribution is significantly related to only one firm success ratio which, interestingly, is ROS. Finally, logistics total cost minimization is significantly and positively related to ROA and ROS although at relatively low levels. Interestingly, minimum total costs is most strongly related to the margin-oriented ROS. In general, all of these results parallel the earlier CEO-rated importance rankings in that logistics demand-management capabilities are most strongly related to firm financial success.

As discussed in the methodology section, stepwise regression was also used to identify the “unique” impacts of logistics capabilities on the same five financial success measures. Table IV shows that all five logistics models are statistically significant at the $p \leq 0.01$ level. In general, the models “explain” between 17 and 29 per cent of the dependent financial variable variance (i.e. R^2), depending on the specific model under consideration. The ROI growth model in Table IV has the highest R^2 .

In terms of which independent variables explain the most significant and unique variance, logistics customer service, and to a lesser degree logistics quality, dominate the results. In contrast, channel distribution and logistics total cost minimization do not remain in any of the five final models as explaining unique variance. Thus, the demand-management capabilities again dominate the supply-management variables in this stepwise analysis.

Business performance dependent variable ^a	Final model F-statistic (p -value)	Final model R^2	Independent standardized variables: parameter estimates β_i of variables entered into final model (p -value)	
ROA	4.97* (0.012)	0.20	Logistics customer service	0.58 (0.01)
			Logistics quality	0.37 (0.10)
ROI	4.90* (0.013)	0.20	Logistics quality	0.45 (0.01)
			Logistics customer service	0.53 (0.12)
ROS	8.06* (0.007)	0.17	Logistics customer service	0.79 (< 0.01)
ROI growth	7.86* (0.001)	0.29	Logistics customer service	0.71 (< 0.01)
			Logistics quality	0.47 (0.04)
ROS growth	5.51* (0.008)	0.22	Logistics customer service	0.67 (< 0.01)
			Logistics quality	0.34 (0.12)

Notes: ^aBusiness performance indicators consist of the performance ratios ROA, ROI, ROS, ROI growth, and ROS growth

*All models are statistically significant at $p \leq 0.01$

Table IV. Stepwise regression models with logistics quality, customer service, channel distribution and total cost as potential independent variables

Summary and discussion

If business processes optimally cut across functions to integrate functional capabilities, then we would expect the functional areas to be significantly related. This indeed is found to be the case in this study. Firms which perform well in one functional area tend to be good performers in all other functions as well. This is summarized in the top panel (A) of managerial summary and overview, Table V.

In general, logistics performance explains between 16 and 25 per cent of the performance variation in other functional areas. From an operational supply perspective, these cross-functional relationships imply functional process excellence. From a customer demand perspective, these results imply that customers can and should expect overall firm excellence rather than compensating tradeoffs between functional areas of a firm.

Table V.
Managerial summary
and overview of results

(A) Relative strength of performance relationships between functional areas and logistics capabilities (only significant relationships are shown) (1 = strongest relationship; * = particularly interesting finding) *1. Logistics customer service and production 2. Overall logistics and marketing 3. Logistics channel of distribution and marketing 4. Overall logistics and production 5. New product development is about equally related with overall logistics, marketing, and production 6. Logistics quality and production 7. Logistics customer service and new product development 8. All other logistics capabilities are about equally related with new product development *9. Marketing and production 10. Logistics total cost minimization and marketing *11. Logistics customer service and marketing *12. Other interface relationships are <i>not</i> statistically significant (e.g. logistics quality and marketing performance; logistics total cost minimization and production performance, etc.) (B) Relative strength of relationships between functional performance and <i>overall</i> firm performance (1 = strongest relationship; * = particularly interesting finding) 1. Overall logistics *2. Logistics customer service *3. Logistics quality 4. New product development 5. Production 6. Logistics total cost minimization is <i>directly</i> related to two overall firm performance measures but has no <i>unique</i> impact 7. Logistics channel of distribution is <i>directly</i> related to one overall firm performance measure but has no <i>unique</i> impact *8. Marketing in this industry is not significantly related to any of the five overall firm performance measures used (ROI, ROA, ROS, ROI growth, and ROS growth) (C) CEO-rated importance of functional areas versus logistics capabilities (1 = most important; * = particularly interesting finding) *1. Logistics quality *2. Logistics customer service 3. Overall logistics 4. Production 5. Logistics channel of distribution 6. Marketing 7. New product development *8. Logistics total cost minimization

The strongest *functional* performance relationships found are between overall logistics and marketing (No. 2 in panel A, Table V) followed by overall logistics and production (No. 4 in panel A, Table V). In contrast, the direct relationships between marketing and production are at a lower level (No. 9 in panel A of managerial summary Table V). These findings support and are consistent with the Introduction's viewpoint that logistics has the primary and strongest interface and boundary spanning role with both functions of the firm (i.e. Figure 2). As such, some would argue that logistics should have the primary role in co-ordinating and integrating all functional supply chain capabilities[10].

The performance analyses involving individual specific logistics capabilities and functional areas also provide additional corroborative support and interesting insights. The strongest function/capability performance relationship found is between logistics customer service and production, and to a lesser degree, logistics quality and again production (top panel A, Table V). Although customer service has had more attention in the logistics literature for marketing performance, e.g. [5,13,14,21], the present research shows that it is also extremely important for production performance. Perhaps these strong production relationships are understandable, since production depends on logistics service for the provision of components and material inputs. Likewise, logistics service flexibility is important for shortages, work stoppage and other emergencies. Finally, in the current era, poor logistics quality, such as late arrivals, can have serious deleterious effects on JIT-type production, even shutting down the line in worst case circumstances.

Logistics customer service is also significantly related to marketing performance, although interestingly, logistics quality is not. Logistics customer service has been positioned in the literature as a major demand interface with the marketing function and as part of the marketing mix[11,13,14]. As such, the present findings demonstrate that customer service has significant performance impacts on overall marketing performance. However, it is important to temper these results with the reminder that this is primarily a CEO perception study. Perceptions and operational realities can be two different things when tapping into CEO viewpoints. Nevertheless CEO perceptions can be quite useful managerial indicators of the best selling route for logistics capabilities, as will be discussed below.

In general, these customer service and quality results are interesting and gratifying since, as outlined in the introduction, both are viewed as representing major demand interfaces of logistics with both production and marketing (e.g. Figure 2). The present results indicate that these demand interfaces have major consequences for interfunctional performance and process integration. In essence, the customer service and logistics quality interfaces represent integrative aspects of logistics boundary spanning.

From a logistics supply perspective, both channel distribution and logistics *total* cost minimization are found to be significantly related to marketing performance. Channel distribution is shown to be the logistics capability variable having the strongest relationship to marketing performance (Table V, panel A, No. 3). This channel advantage as well as other high ranking examples in managerial summary Table V (e.g. logistics customer service and production is No. 1 in panel A), can be leveraged by logisticians. It has been stated that logisticians must learn to “speak the language” of top management[24] and to enlist the support of allies to their positions. Table V and its method are indicative of logistics advantages which are consistent with top management perceptions and likely to find favour with specific functional managers within this industry. Basically, these are the “easiest selling points” within these internal organizations.

As defined and discussed in the introduction (e.g. Figure 2), channel distribution can conceptually be thought of as the supply “place” component of the marketing mix with its four Ps of place, product, promotion, and price[5,21,23]. In turn, customer service can be thought of as the demand “place” component of the marketing mix (Figure 2). In fact, each of these other marketing mix Ps can also be thought of as having both supply and demand interface components. For logistics’ supply contribution to “place”, the present finding of a particularly strong channel relationship with marketing performance supports this conceptualization. In turn, customer service is shown to also have strong demand interfaces with functional areas. This explains the apparent contradiction as to both channels, e.g. [21,23] and customer service e.g. [13,14] representing “place” interfaces with marketing. Thus, logistics does not have solely one demand (customer service) interface with other functional areas but other major supply and demand interfaces as well (e.g. channels, quality, price, etc.).

Next, the significant relationship found between logistics total cost minimization and marketing performance possibly reflects the present ascent and current utilization of the “total cost concept” (TCC) by marketing decision makers. For example, TCC relates to price and service advantages, both of which directly impact marketers’ performance. In turn, CEOs may rate TCC lowest in importance because they only see the budget information of functional “silos”.

In contrast to marketers’ performance, total cost minimization is not found to be significantly related to production performance (nor are channels). Possibly the function of production continues to stress resource costs such as materials and labour rather than correctly adopting the total cost minimization concept. The latter is preferable since it explicitly requires cross-functional cost tradeoffs for system optimization, rather than risking functional suboptimization.

The present research is somewhat unique in that it includes the variable of new product development in the performance relationship tests. New product development performance is found to be uniformly and consistently related to all other functional areas. Similarly, it is uniformly related to all individual logistics capabilities, although at relatively low levels. As conceptualized in the Introduction, the picture that emerges for new product development is that it touches all functional areas and all logistics capabilities, yet it is not dominated by any particular area. As such, new product development can be viewed as a “platform” for other functional areas as depicted in Figure 2.

The stepwise regression analyses address the important question as to which logistics capabilities have the greatest *unique* impact on overall firm performance. Basically, stepwise regression investigates the impact of each potential logistics capability variable after statistically holding constant the effects of all other logistics capability variables. Five firm financial performance variables are used as individual dependent variables in five separate stepwise regression models.

The stepwise models show that the demand-management variables are more important for distinguishing successful firms than the supply-management

variables (e.g. summary Table V, panel B, No. 2 and No. 3). Specifically, logistics customer service and quality are uniquely and significantly related to overall firm financial performance, while channel distribution and total logistics cost minimization are not uniquely related. These results parallel the CEO-rated importance rankings, in that the demand-management variables of customer service and quality were rated by CEOs as more important for firm success (e.g. summary Table V, panel C, No. 1 and No. 2). Perhaps supply-management variables are only order qualifiers or the “minimum ante to play”, while demand-management capabilities are the true order winners. However, it is also possible that in upscale, fast moving consumer goods or computer industries, supply capabilities and supply innovation may become more critical success factors. Furthermore, of the two demand capabilities, customer service has a somewhat greater impact on overall firm financial performance. Nevertheless the stepwise regression results imply that customer service and logistics quality are separate and distinct concepts since they both have unique impacts on overall business performance.

Finally, overall logistics and new product development (NPD) are found to be the most important functions for firm financial success (e.g. summary Table V, panel B, No. 1 and No. 4). NPD is most strongly related to ROI-based performance measures. Interestingly, top management rated NPD as the least important *functional* area for firm success (summary, Table V, panel C, No. 7), yet it is more strongly related to firm performance than marketing and production. Perhaps the latter two are also merely order qualifiers while NPD and logistics represent true order winners and sustainable competitive advantages. However, most of the other CEO importance rankings (i.e. for overall logistics and its specific demand capabilities in panel C of Table V) are in line with the identified financial performance relationships (panel B in summary Table V).

Conclusions and implications

The present research addresses performance results from both interfunctional process integration and specific logistics interface capabilities. In terms of cross-functional process excellence, the study findings establish that firms which perform well in one functional area tend to perform well in all other functional areas. This implies that excellence solely in one functional area is not likely to be a basis for competitive advantage. Rather, process integration across functional areas becomes a source of competitive advantage for better performing firms.

These results may also imply that an organizational culture of cross-functional excellence creates performance synergies across functional boundaries. Indeed, as conceptualized in the Introduction and in model Figure 2, functional boundaries need to be made flexible and virtually transparent in the pursuit of cross-functional excellence. Managerially, this implies co-operation, information sharing, co-ordination and joint planning across organizational boundaries. Specific tools and mechanisms to achieve these goals can be generally found in organizational behaviour management

literature and specifically for logistics in the materials referenced at the end of this article (e.g. [25]). Some of these practices, such as top management support and better cross-functional communications, are not particularly esoteric and may be relatively easy to adopt.

The strong boundary-spanning role found for logistics with both production and marketing implies that logistics can be used as a vehicle for cross-functional integration, a nexus of communication and co-ordination, and for better system performance. More generally, it gives new meaning to the view that logistics can be a competitive weapon by identifying and supporting logistics' boundary spanning role as a major and unique source of competitive advantage.

Specific logistics interface capabilities identify the sources of boundary-spanning advantages. Although customer service has received considerable attention in the logistics literature as a demand interface with marketing, the particularly strong relationships found with production suggests a relatively neglected area of customer service advantage. Logistics quality provides a second source of interface advantage. Both would appear particularly appropriate in the current demand-pull or JIT-type environment.

The extremely strong relationship found between channel distribution and marketing performance implies that logistics channel performance also has major impacts on marketing performance. While some logistics literature identifies customer service as the interface with marketing performance, e.g. [13,14], other logistics literature, e.g. [21,23], identifies channel distribution as the interface with marketing. It appears that both viewpoints are correct. Logistics has both demand-management and supply-management interfaces with marketing performance as well as with other functional areas (see Figure 2). Similarly, the significant relationship found with total cost minimization may imply marketing acceptance of integrated supply chain management. Specifically, these findings support those who contend that logistics and marketing should both be involved in channel decisions. More broadly, it supports the view that channels should be designed and managed jointly by marketing and logistics personnel rather than by fragmented channel efforts.

Although cross-functional and interface performance relationships are important and interesting, many business executives are also interested in the "bottom line". In terms of which functional areas have the greatest impact on financial performance and growth, logistics followed by NPD were identified as having by far the strongest impacts. Managerial summary Table V provides an executive overview of these and other study findings and highlights some of the most interesting findings. The previous summary and discussion section also summarizes and elaborates on overall study results. The strong and significant logistics relationships with all five profitability and growth measures empirically supports what some logistics executives have been saying all along. However, the NPD finding is interesting because the CEO-importance rankings placed it last in *functional* importance after both production and marketing (see summary Table V). NPD seems to ideally relate to all functional areas about equally, and thus, can be thought of as a "platform" as depicted in Figure 2.

However, several cautionary notes and observations should be sounded with these findings. First, results appear to be sensitive to the financial measure selected. For example, previous research has found marketing to be an important source of market share gains which is not tested here. Second, functional competences, such as marketing and production, may be order qualifiers but not order winners. Strategically, they may be necessary functional competences, but do not provide the firm with that extra competitive edge which shows up in bottom-line profitability and growth. In contrast, logistics and NPD may provide that extra edge.

The stepwise regressions demonstrate that only the demand-management interface capabilities of customer service and logistics quality uniquely “explain” business profitability and growth. Again, these would seem to be the order winners or the extra competitive edge that shows up in a company’s profits and growth. This would also seem to imply that management should concentrate on demand-management capabilities if supply capabilities are competitive or, more importantly, meet customer expectations. However, three important observations should be made. First, this conclusion may again not totally apply for upscale, fast moving products. Second, customer service and logistics quality are separate practices as evidenced by their simultaneous appearance in the majority of stepwise models (as explaining unique variance). Consequently, they both deserve managerial attention to become company best practices as defined in the Introduction. Third, customer service and logistics quality are not synonymous with marketing. In terms of process integration, the previous results showed that they had much stronger relationships with production than marketing, which supports logistics’ strong interface role. In total, this again implies that true competitive advantages emanate from leveraging cross-functional logistics boundary spanning rather than stressing functional (sub) optimization. It also implies that boundary-spanning advantages may be found where least expected and, therefore, least imitable by competitors.

Future research might replicate this study in other industries. The research team firmly believes that a sufficient number of single industry investigations must be conducted before industry effects can be isolated. For example, by concentrating on one industry, differences in industry effects are controlled and do not confound or obfuscate findings. Nevertheless, the resulting limitation on generalizability must be considered. Other logistics boundary-spanning research could also be carried out in other areas besides marketing, production and new product development. For example, important interfaces are likely to exist with purchasing, management information systems and technology, human resource planning and law. In fact, efforts towards a unified conceptualization and theory of boundary-spanning interfaces must remain tentative until all of this research is accomplished.

References

1. Langley, C.J. Jr and Holcomb, M.C., “Creating logistics customer value”, *Journal of Business Logistics*, Vol. 13 No. 2, 1992, pp. 1-27.

2. Hammer, M., "Reengineering work: don't automate, obliterate", *Harvard Business Review*, No. 4, July-August 1990, pp. 104-12.
3. Shapiro, B., Rangan, V.K. and Sviokla, J.J., "Staple yourself to an order", *Harvard Business Review*, No. 4, July-August 1992, pp. 113-22.
4. Ballou, R.H., *Business Logistics Management*, 3rd ed., Prentice-Hall, Englewood Cliffs, NJ, 1992, p. 25.
5. Lynagh, P.M. and Poist, R.F., "Assigning organisational responsibility for interface activities: an analysis of PD and marketing manager preferences", *International Journal of Physical Distribution & Materials Management*, Vol. 14 No. 6, 1984, pp. 34-46.
6. Bowersox, D.J. and Closs, D.J., *Logistical Management*, 4th ed., McGraw-Hill, Hightstown, NJ, 1996.
7. Bowersox, D.J., Morash, E.A. and Daugherty, P.J., "Boundary spanning in channels of distribution", in Kahn, K.D. (Ed.), *Developments in Marketing Science: Proceedings of the 1988 Academy of Marketing Science Annual Conference*, Vol. 11, April 1988, pp. 470-4.
8. Leifer, R.P. and Delbecq, A., "Organization/environment interchange: a model of boundary spanning activity", *Academy of Management Review*, Vol. 3, January 1978, pp. 40-50.
9. Schary, P.B., "Customer service as a system process", in Lusch, R.F. and Zinszer, P.H. (Eds), *Contemporary Issues in Marketing Channels*, University of Oklahoma Distribution Research Program, Norman, OK, 1979, pp. 165-76.
10. Morash, E.A., "Customer service, channel separation and transportation intermediaries", *Journal of Business Logistics*, Vol. 7 No. 1, 1986, pp. 89-107.
11. Lambert, D.M. and Stock, J.R., *Strategic Logistics Management*, 3rd ed., Dow-Jones Irwin, Homewood, IL, 1993.
12. Morash, E.A., "On the use of logistics strategies to promote demand", *The Logistics and Transportation Review*, Vol. 26 No. 1, March 1990, pp. 53-75.
13. Lambert, D.M. and Harrington, T.C., "Establishing customer service strategies within the marketing mix: more empirical evidence", *Journal of Business Logistics*, Vol. 10 No. 2, 1989, pp. 44-60.
14. Sterling, J.U. and Lambert, D.M., "Establishing customer service strategies within the marketing mix", *Journal of Business Logistics*, Vol. 8 No. 1, 1987, pp. 1-30.
15. Stock, J.R. and Lambert, D.M., "Becoming a 'world class' company with logistics service quality", *International Journal of Logistics Management*, Vol. 3 No. 1, 1992, pp. 73-81.
16. Christopher, M., Schary, P. and Skjott-Larsen, T., *Customer Service and Distribution Strategy*, John Wiley & Sons, New York, NY, 1979.
17. Christopher M., *Logistics and Supply Chain Management*, Irwin Professional, Burr Ridge, IL, 1994.
18. LaLonde, B.J. and Zinszer, P.H., *Customer Service: Meaning and Measurement*, National Council of Physical Distribution Management, Chicago, IL, 1976.
19. LaLonde, B.J., Cooper, M.C. and Noordewier, T.G., *Customer Service: A Management Perspective*, Council of Logistics Management, Oak Brook, IL, 1988.
20. Ozment, J. and Morash, E.A., "The augmented service offering for perceived and actual service quality", *Journal of the Academy of Marketing Science*, Vol. 22 No. 4, 1994, pp. 352-63.
21. Lynagh, P.M. and Poist, R.F., "Managing physical distribution/marketing interface activities: co-operation or conflict?", *Transportation Journal*, Vol. 23 No. 3, 1984, pp. 36-43.
22. Mentzer, J.T. and Konrad, B.P., "An efficiency/effectiveness approach to logistics performance analysis", *Journal of Business Logistics*, Vol. 12 No. 1, 1991, pp. 33-61.
23. Bowersox, D.J. and Morash, E.A., "The integration of marketing flows in channels of distribution", *European Journal of Marketing*, Vol. 23 No. 2, 1989, pp. 58-67.
24. Novack, R.A., Rinehart, L.M. and Langley, J. Jr, *Creating Logistics Value: Themes for the Future*, Council of Logistics Management, Oak Brook, IL, 1995.
25. Murphy, P.R. and Poist, R.F., "The logistics marketing interface: techniques for enhancing co-operation", *Transportation Journal*, Vol. 32 No. 2, Winter 1992, pp. 14-23.

Appendix. Functional area capabilities and their definitions

Boundary-
spanning
interfaces

Functional area capability	Capability definitions and measures
(A) Production	
1. Production customization	The ability to produce products characterized by numerous features, options, sizes and/or colours to meet specific customer specifications
2. Production quality	The ability to manufacture a product whose operating characteristics meet established performance standards
3. Low cost production	The ability to minimize the cost of production (inclusive of labour, materials, and operating costs) through efficient operations, process technology, and/or scale economies
4. Production flexibility	The ability to rapidly adjust capacity to accelerate production in response to changes in customer demand
5. Production lead time	The ability to reduce the time it takes to manufacture products
6. Process flexibility	The ability to produce low quantities of product, cost efficiently so that product mix changes are easily accommodated
7. Product reliability	The ability to maximize the time to product failure or malfunction
8. Product durability	The ability to maximize the time to product replacement
(B) Marketing	
1. Market segmentation	The ability to identify promising target markets and to select the best ones for marketing
2. Brand image	The ability to create a positive or favourable image in customers' minds when they hear the product's brand name
3. Advertising and promotion	The ability to create effective advertising and/or promotional campaigns
4. Competitive pricing	The ability to offer a lower product price than direct competitors
5. Selling proficiency	The ability to move products successfully through personal selling activities
6. Company reputation	The ability to create a positive or favourable image in customers' minds when they hear the company's name
7. Low price	The ability to offer one of the lowest product prices
8. Broad product line	The ability to provide a comprehensive set of related items within a given product line offering
(C) New product development	
1. Design innovation	The ability to design a product with capabilities, features, styling, etc., which is superior to competing products
2. New product development	The ability to develop new products for existing markets
3. Original development	The ability to develop original (i.e. "new-to-the-world") products which create entirely new markets
4. Development cycle time	The ability to minimize the time it takes to develop new products
5. New product innovation	The ability rapidly to introduce large numbers of product improvements/variations or completely new products
6. Technological innovation	The ability to engage in new product development involving major advances in product technology
7. Product improvement	The ability to develop and refine existing products further

369